





From Characters to Meaning: Core Techniques in Natural Language Processing

DSA0308 – Natural Language Programming

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INTRODUCTION

What is NLP?
Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language.

Goal:
Understand human language
Process text and speech
Generate meaningful responses

Examples
ChatGPT
Google Translate
Siri & Alexa
Grammarly
Spam Detection




NLP Pipeline

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graph TD
    A[Raw Text] --> B[Text Preprocessing]
    B --> C[Tokenization]
    C --> D[Stop Word Removal]
    D --> E[Stemming/Lemmatization]
    E --> F[Feature Extraction]
    F --> G[Machine Learning / Deep Learning]
    G --> H[Prediction or Response]
  
```




Text Preprocessing

Techniques

- Lowercase conversion
- Remove punctuation
- Remove numbers
- Remove special characters
- Remove extra spaces

Example
Original:
I LOVE NLP!!! 😊
Processed:
i love nlp




Tokenization

Tokenization splits text into smaller units called **tokens**.

Example
Sentence:
I love Natural Language Processing

Tokens:
["I", "love", "Natural", "Language", "Processing"]

Types

- Word Tokenization
- Sentence Tokenization
- Character Tokenization




Stemming vs Lemmatization

Stemming Cuts words to their root. Examples playing → play studies → studi running → run Fast but less accurate.	Lemmatization Uses dictionary meaning. Examples running → run better → good studies → study More accurate than stemming.
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Applications



- Chatbots
- Machine Translation
- Speech Recognition
- Text Summarization
- Sentiment Analysis
- Spam Detection
- Fake News Detection
- Medical Text Analysis



Future of NLP



- Large Language Models (LLMs)
- Multilingual AI
- Conversational AI
- AI Assistants
- Healthcare
- Education
- Legal Document Analysis
- Autonomous Systems

THANK YOU

